Introduction (1)

Period: 6
Group Members: Ethan Tran
Group Name: Project Kairos
Project Title: 3D Kairos

Description (2 & 4)

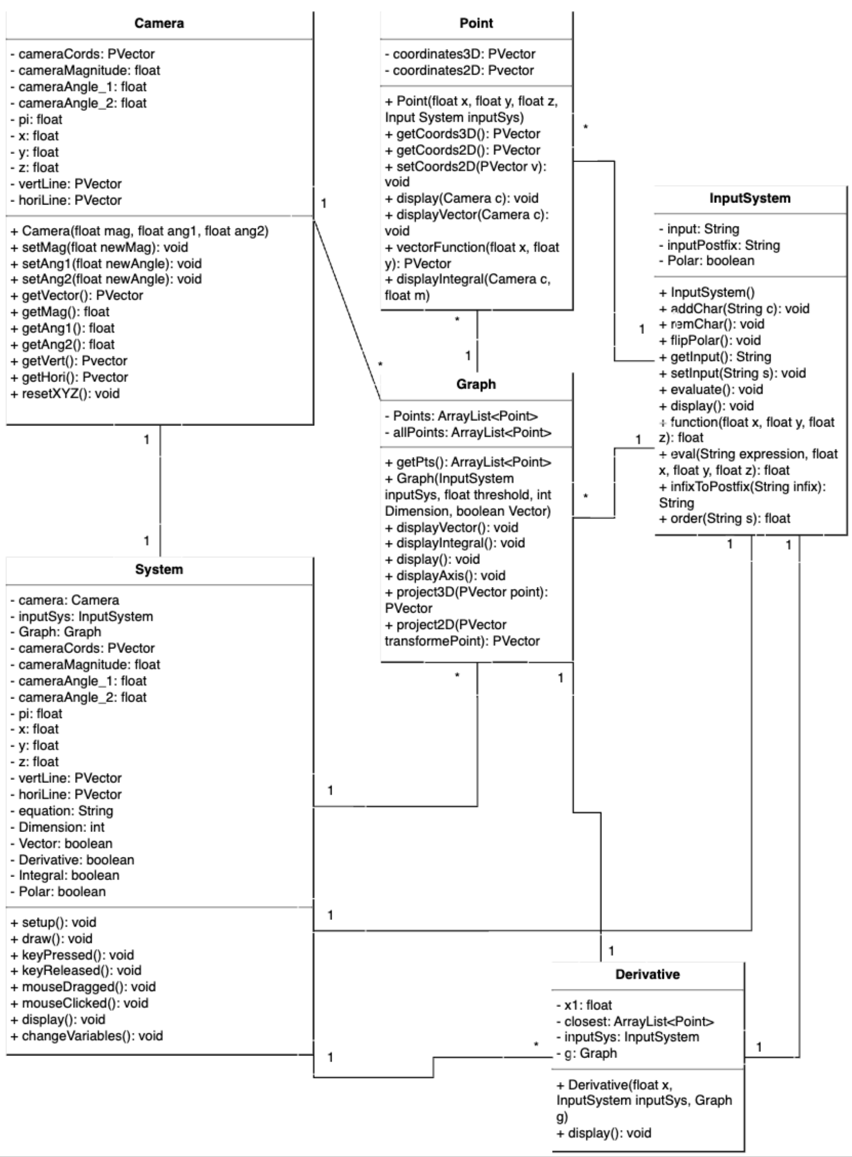
Project Overview: The project that I will be working on is a 3-dimensional graph visualizer, similar to 3D Desmos. In essence, the program will allow the user to graph 3D Functions by inputting equations of their choice and move around the space to view their creation from a variety of perspectives.

Functionalities and How They Work:

- Visualization of 3D Space: The image displayed will be a projection of a 3D space onto the 2D screen in order to simulate said 3D space.
  - How does it work?: The 3D space is automatically active (you can toggle between 3D and 2D by pressing the 'D' key). The graph of the current function is always shown.
- Graphing Capabilities: The program will be able to plot points along a graph in both 2-dimensional and 3-dimensional space, working in tandem with the visualizer.
  - How does it work?: Like the visualizer, this is always active and does not need anything to be turned on. To utilize the graphing capabilities, the user must interact with the Input System (below) and enter their function of choice.
- Vector Fields: In the 2-dimensional visualizer, the program will allow the user to plot Vector Fields in two-dimensional space (technically creating a 4-Dimensional space)
  - How does it work?: When in 2D mode, Vector Field can be toggled by the 'V' key. It will take the current function and display arrows accordingly.
- Integrals: Many functions are non-elementary (cannot be written with typical functions such as logarithms, addition, multiplication, subtraction, exponentiation, division, trigonometric functions, etc.) and thus require integrals (signed area under a curve) to evaluate. I plan on implementing the integral to graph higher-order functions.
  - How does it work?: When in 2D mode, the Integral function can be used by typing "int(FUNCTION)" which will estimate the area under the curve by using many thin rectangles (Riemann Sums).
- Derivatives: A higher-order function that can be applied to 2D and 3D graphs to find a tangent line or tangent plane.
  - How does it work?: The Derivative function can be used by typing "d/dx(FUNCTION,x)" which will find the tangent plane to the graph at a specific point (x).
- Translation, Rotation, Scaling: These transformations allow us to explore the graph much better.
  - How does it work?: The transformations can be activated by the use of several keys. 'WASD' for transformations, Arrows Keys to Rotate the Camera, and 'Q' and 'E' keys to zoom in and out.

- Polar Coordinates: An alternate way to graph 3D and 2D graphs that centers around circles, compared to rectangular coordinates.
  - <u>How does it work?</u>: When in any mode, Polar mode can be toggled by the ‘P’ key. The same variables (x,y, and z) will be used; however, x becomes r (radius), y becomes theta (azimuthal angle), and z becomes phi (polar angle).
- Input System: A system that can take in function inputs from the user and create a new graph.
  - <u>How does it work?</u>: By clicking the ‘return’ key, the user can activate or deactivate input mode. When input mode is active, the user can type functions (and visually see what they are typing) without fear of activating another mode above. Once the function is complete, clicking the ‘return’ key again will send the function to be graphed.

UML Diagram (3)



Functionalities / Issues (5)

- Current Functionalities:
  - Complete 2D and 3D Visualization through projection (and additional optimizations)
    - With angles and planes
    - With unit vectors
  - Graphing Capabilities: The user can manually input their desired function (not through an input system yet) into the grapher and see it with the 3D visualization.
  - Improved comprehensibility of my code (Split the single class into multiple different classes)
  - Camera Rotation and Scaling with ‘Q’, ‘E’, and WASD keys
  - Function Input System (Number keys, ‘.’, ‘+’, ‘-’, ‘*’, ‘/’, ‘x’, ‘y’, ‘z’)
    - Can type in specific keys to create functions in a viewable input area.
  - Vector Fields: The user can enter vector field mode by pressing the ‘v’ key, which will show a vector field of the given equation.
  - Derivatives (‘,’): The user can find the line tangent to the given function at specific x-values by clicking or dragging the mouse when Derivative mode is active.
  - Integrals (‘i’): The user can see a Riemann Sum representation of the signed area under the curve. Text at the top left of the screen is an estimate of the area under the curve.
  - Polar Coordinates (‘p’): The user can utilize Polar Coordinates instead of Rectangular coordinates.
- Unfinished Functionalities:
  - Camera Translation: Translation would require a complete reworking of the camera and projection system, as the current iteration relies on the origin being the focus point.
- Bugs and Issues
  - Issue with camera rotation inverting the shape/direction of rotation: I solved this problem by bashing all of the combinations of angles and radii to individually solve every problem with the camera.
  - Issue with slow processing: A large quantity of points requires much time to completely load (especially with larger thresholds). Thus, I revamped the projector with a new system in an attempt to simplify the whole process and get a faster time complexity.
  - Issue with slow input graphing: The Input System used to find the equation for the input every time the function is graphed, which was incredibly inefficient, so I changed it to only perform this action once.
  - Multiple issues with graphing vectors: Fixed issue with specific angles not working, line segments in the wrong directions, and errors when dividing by 0.

Log (6)

Functionality	Creator	Date
3D Visualization with Angles	Ethan Tran	05/14
3D Visualization with Unit Vectors	Ethan Tran	05/18
Graphing Capabilities	Ethan Tran	05/16
Camera Rotation	Ethan Tran	05/15 and 05/19 (after switching methods)
More Classes!	Ethan Tran	05/18
Input System	Ethan Tran	05/22
Rectangular Graph Class	Ethan Tran	05/23
Vector Fields	Ethan Tran	05/25
2nd Dimension	Ethan Tran	05/24
Fixed Rotation Bug	Ethan Tran	05/29
Integrals	Ethan Tran	05/30
Polar Coordinates	Ethan Tran	05/30
Enhanced Derivatives	Ethan Tran	05/31
UI	Ethan Tran	05/31
Axes	Ethan Tran	05/31
Optimization	Ethan Tran	05/31